Sequential Monte Carlo^2 for Bayesian Filtering and Stochastic Optimal Control of a Physiological SDE

Lecture notes on the FSA-v2 model

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Abstract

These notes present the mathematical theory underlying a research code base that combines particle filtering and stochastic optimal control on a three-dimensional physiological stochastic differential equation. We define the FSA-v2 model in its G1-reparametrized form, prove that all 30 static parameters are locally identifiable from a one-day window of four-channel observations by analytical Fisher-information-matrix calculation, and establish exponential ergodicity of the SDE on its operating domain via a quadratic Lyapunov function. We then develop the bootstrap particle filter and the SMC^2 algorithm of Chopin, Jacob and Papaspiliopoulos, recast optimal scheduling of training stimulus as posterior inference under a tempered cost likelihood, and close the loop in a model-predictive controller that interleaves a rolling SMC^2 filter with a posterior-mean control SMC^2 over an RBF schedule. The receding-horizon section is presented in both abstract mathematical and code-level form, and includes a linear–quadratic–Gaussian / differential-Riccati alternative that is sketched in full but flagged as the principal piece of remaining analytical work.

The audience is graduate students in applied mathematics, mathematical engineering, applied stochastics, or process engineering. Familiarity with Itô stochastic differential equations, Bayesian filtering, and basic optimal control at the level of a one-semester course is assumed; the particle-filtering and tempered-SMC machinery is built up from first principles.

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1 Introduction

A human athlete is, from the point of view of an applied mathematician, a non-linear stochastic dynamical system whose state is only ever observed through a handful of noisy proxies. Heart rate, sleep duration, perceived stress, and step counts are all loosely informative about the latent fitness, fatigue, and autonomic condition of the body, but none of them directly measures any of these quantities. The classical Banister model of training response (Banister, 1991) treats the athlete as a deterministic two-compartment system; the present notes work with a stochastic three-compartment refinement, denoted FSA-v2, in which fitness, fatigue, and an autonomic amplitude evolve as a coupled Itô stochastic differential equation driven by a single exogenous control — the daily training stimulus $\Phi(t)$.

Two questions follow naturally:

1. *Filtering.* Given a stream of multi-channel daily observations, what is the joint posterior distribution over the latent state (B, F, A) and the static parameters θ of the SDE?

2. *Control.* Given the same posterior, what schedule $\Phi : [0, T] \to [0, \Phi_{\max}]$ maximises a chosen functional of the latent trajectory, subject to a soft constraint that fatigue $F(t)$ should not exceed a physiological ceiling $F_{\max}$?

Both questions sit at the intersection of Bayesian inference and stochastic optimal control. They are usually treated as separate disciplines; one of the contributions of the underlying code base is to express both within a single sequential Monte Carlo framework. The filter is built from a bootstrap particle filter, lifted to a sampler over static parameters by the SMC^2 algorithm (Chopin, Jacob and Papaspiliopoulos, 2013) and operated in a rolling-window mode. The controller is built by recasting the cost-minimisation problem as posterior inference under a tempered cost likelihood, in the spirit of Toussaint (2009) and the more general control-as-inference literature (Kappen, 2005; Levine, 2018). The two machines are then composed in a model-predictive controller that re-plans the next twelve hours of training stimulus from the most recent posterior.

These notes are organised so that the mathematical content can be read linearly. Section 2 defines the FSA-v2 SDE and observation model in their G1-reparametrized form. Section 3 carries out the Fisher-information-matrix analysis: an analytical derivation of the three rank-deficient parameter pairs of the original parametrization, the reparametrization that absorbs them, and the empirical evidence that the resulting model is fully identifiable at one-day windows. Section 4 establishes stability of the SDE: Jacobian-based local analysis of the deterministic skeleton, the Stuart–Landau bifurcation locus, and a quadratic Lyapunov function satisfying a Foster–Lyapunov drift inequality, from which exponential ergodicity follows. Section 5 develops the bootstrap particle filter. The SMC^2 outer loop and its rolling-window variant are presented in Section 6. Section 7 formulates the control problem as a posterior-inference problem. Section 8 is the extended core of the document: it develops the receding-horizon principle, describes the FSA-v2 model-predictive controller in both mathematical and implementation terms, and sketches a linear–quadratic–Gaussian approximation that could be added as a cheap baseline. Section 9 reports the two headline empirical results of the code base: an open-loop gain of +28% in mean autonomic amplitude over a constant-stimulus baseline on an 84-day horizon, and a closed-loop gain of +17% in a fourteen-day rolling controller. Section 10 closes by reviewing the identifiability and stability analyses just completed, and by identifying the LQG / Riccati alternative as the principal piece of remaining theoretical work.

2 The FSA-v2 SDE model

2.1 State, drift, and diffusion

Let $(B(t), F(t), A(t))$ denote the latent state of a single individual at time $t \geq 0$, measured in days. The three components are interpreted as

- $B(t) \in [0, 1]$ – a normalised *fitness* variable; the Banister chronic compartment;
- $F(t) \in [0, \infty)$ – an unnormalised *fatigue* variable; the Banister acute compartment;
- $A(t) \in [0, \infty)$ – the *amplitude* of an autonomic / circadian Stuart–Landau oscillator.

The single exogenous control input is the (non-negative) training-stimulus rate $\Phi(t) \in [0, \Phi_{\max}]$, expressed in arbitrary daily TRIMP-like units.

The model is presented in the *G1-reparametrized* form. As we will see in Section 3, an earlier form of this model contained three pairs of parameters that the four-channel observation model could not separate. The G1 form rotates each pair into one strongly identified parameter and one residual; the drift formulas below are mathematically equivalent to the older form at a chosen operating point $(B_{\text{typ}}, F_{\text{typ}}, A_{\text{typ}})$ that we now fix.

Definition 1 (Operating point)**.** We work throughout at the reference operating point

$$A_{\text{typ}} = 0.10, \qquad F_{\text{typ}} = 0.20, \qquad \Phi_{\text{typ}} = 1.0. \tag{1}$$

This is the regime of a de-trained athlete training at the canonical Banister default load. The reparametrization is centred at this point, and prior beliefs on the residual parameters encode "the system is typically near here."

The drift coefficients are

$$\mu(B, F) = \mu_0 + \mu_B B - \mu_F F - \mu_{FF} (F - F_{\text{typ}})^2, \tag{2}$$

$$\mathrm{d}B = \left[\kappa_B \cdot \frac{1 + \varepsilon_A A}{1 + \varepsilon_A A_{\text{typ}}} \Phi(t) - \frac{B}{\tau_B} \right] \mathrm{d}t + \sigma_B \sqrt{B(1 - B)}\, \mathrm{d}W_B, \tag{3}$$

$$\mathrm{d}F = \left[\kappa_F \Phi(t) - \frac{1 + \lambda_A A}{1 + \lambda_A A_{\text{typ}}} \frac{F}{\tau_F} \right] \mathrm{d}t + \sigma_F \sqrt{F}\, \mathrm{d}W_F, \tag{4}$$

$$\mathrm{d}A = \left[\mu(B, F)\, A - \eta\, A^3 \right] \mathrm{d}t + \sigma_A \sqrt{A}\, \mathrm{d}W_A, \tag{5}$$

where W_B, W_F, W_A are independent scalar Brownian motions.

The diffusions in (3)–(4) are unchanged from the pre-G1 model: a Jacobi diffusion on $[0, 1]$ for B (Forman and Sørensen, 2008) whose volatility vanishes at both endpoints, and square-root (CIR-type, Cox–Ingersoll–Ross 1985) diffusions for F and A whose volatility vanishes at the lower boundary.

Reading the residual factors. The two ratios $(1 + \varepsilon_A A)/(1 + \varepsilon_A A_{\text{typ}})$ and $(1 + \lambda_A A)/(1 + \lambda_A A_{\text{typ}})$ each equal one when $A = A_{\text{typ}}$. So at the operating point, and for the constant control $\Phi \equiv \Phi_{\text{typ}}$, the drift collapses to

$$\dot{B} = \kappa_B \Phi - B/\tau_B, \qquad \dot{F} = \kappa_F \Phi - F/\tau_F, \qquad \mu = \mu_0 + \mu_B B - \mu_F F. \tag{6}$$

That is, the strongly identified parameters κ_B, τ_F, μ_F play exactly the role they would in a Banister model linearised at $(F_{\text{typ}}, A_{\text{typ}})$. The residuals $\varepsilon_A, \lambda_A, \mu_{FF}$ control departures from the operating point, and they do so via terms that vanish at the operating point itself; this is what makes them weakly identified at one-day windows and consequently candidates for tighter priors.

2.2 Truth parameters

Table 1 lists the truth parameters of the G1-reparametrized model. Compared to the pre-G1 form, three values change: $\kappa_B = 0.012 \cdot (1 + \varepsilon_A A_{\text{typ}}) = 0.01248$, $\tau_F = 7.0/(1 + \lambda_A A_{\text{typ}}) = 6.\overline{36}\,\text{d}$, and $\mu_F = 0.10 + 2F_{\text{typ}}\mu_{FF} = 0.26$, with a small additive shift $\mu_0 \mapsto \mu_0 + \mu_{FF}F_{\text{typ}}^2 = 0.036$ absorbing the constant from completing the square in (2).

Group	Symbol	Value	Meaning
Banister timescales	τ_B	42.000	chronic time constant (d)
	τ_F	6.364	effective acute time constant (d)
Banister gains	κ_B	0.01248	effective B-gain at A_{typ}
	κ_F	0.030	F-gain per unit Φ
A-coupling (residual)	ε_A	0.40	residual B-gain boost beyond A_{typ}
	λ_A	1.00	residual F-clearance boost beyond A_{typ}
Stuart–Landau	μ_0	0.036	growth-rate intercept
	μ_B	0.30	fitness raises μ
	μ_F	0.26	local slope of μ at F_{typ}
	μ_{FF}	0.40	residual curvature of μ in F
	η	0.20	cubic damping
Diffusion	σ_B	0.010	Jacobi noise scale
	σ_F	0.012	CIR noise scale on F
	σ_A	0.020	CIR noise scale on A

Table 1: Truth parameters of the G1-reparametrized FSA-v2 dynamics. Source: `version_2/models/fsa_high_res/_dynamics.py`.

2.3 The training-stimulus envelope

The control variable in the SDE is the instantaneous training-stimulus rate $\Phi(t)$. In practice training is delivered in discrete daily sessions concentrated in the morning, and the simulator expands a daily schedule $(\Phi_d)_{d=1,\dots,N}$ into a high-resolution envelope at fifteen-minute resolution as follows. Let $\tau_{\text{burst}} = 3\,\text{h}$ be a shape parameter, and let $[t_{\text{wake}}, t_{\text{sleep}}] = [7, 23]$ be the daily wake window (in hours). The within-day envelope is

$$e(t) \;\propto\; \begin{cases} (t - t_{\text{wake}})\,\exp\!\big(-(t - t_{\text{wake}})/\tau_{\text{burst}}\big) & t_{\text{wake}} \le t < t_{\text{sleep}}, \\ 0 & \text{otherwise}, \end{cases} \tag{7}$$

which is the standard Gamma($k = 2$) shape, peaking at $t - t_{\text{wake}} = \tau_{\text{burst}}$. The constant of proportionality is fixed so that $\int_{\text{day}} \Phi(t)\,\mathrm{d}t = 24\,\Phi_d$, i.e. the daily integral of the within-day envelope is preserved. The slow Banister equations (3)–(4) therefore see the same daily load whether Φ is delivered as a constant or as a morning-loaded burst, and the fast Stuart–Landau dynamics in A pick up the within-day shape.

2.4 Four-channel observation model

The state (B, F, A) is not directly observed. Instead we receive four discrete-time channels at the fifteen-minute grid:

$$\mathrm{HR}(t) \sim \mathcal{N}\big(\,\mathrm{HR}_{\mathrm{base}} - \kappa_B^{\mathrm{HR}}\, B + \alpha_A^{\mathrm{HR}}\, A + \beta_C^{\mathrm{HR}}\, C(t),\ \sigma_{\mathrm{HR}}^2\big), \qquad \text{(sleep-gated)} \tag{8}$$

$$P(\mathrm{sleep}(t) = 1 \mid A, C) = \sigma\big(\,k_C\, C(t) + k_A\, A - \tilde{c}\,\big), \qquad \text{(always observed)} \tag{9}$$

$$S(t) \sim \mathcal{N}\big(\,S_{\mathrm{base}} + k_F\, F - k_A^S\, A + \beta_C^S\, C(t),\ \sigma_S^2\big), \qquad \text{(wake-gated)} \tag{10}$$

$$\log(\mathrm{steps}(t) + 1) \sim \mathcal{N}\big(\,\mu_0^{\mathrm{step}} + \beta_B^{\mathrm{step}}\, B - \beta_F^{\mathrm{step}}\, F + \beta_A^{\mathrm{step}}\, A + \beta_C^{\mathrm{step}}\, C(t),\ \sigma_{\mathrm{step}}^2\big), \qquad \text{(wake-gated)} \tag{11}$$

where $C(t) = \cos(2\pi t + \phi)$ is a fixed deterministic circadian covariate (not a latent variable). "Sleep-gated" channels are emitted only when the sleep label of (9) is one; "wake-gated" channels only when it is zero. The collection of observation-model parameters $(\mathrm{HR}_{\mathrm{base}}, \kappa_B^{\mathrm{HR}}, \ldots, \sigma_{\mathrm{step}})$ brings the full static parameter vector θ_{dyn} for the joint state-space model to 30 scalar entries. We write $y_{1:n}$ for the multi-channel observation record up to time t_n and $p(y_{1:n} \mid \theta_{\mathrm{dyn}})$ for its marginal likelihood under the model.

The two analyses that an earlier version of these notes flagged as outstanding – Fisher Information rank identifiability and Lyapunov stability – have since been carried out and form Sections 3 and 4 respectively.

3 Identifiability via the Fisher information matrix

A model is no use if its parameters cannot be recovered from the observations: an inference procedure may converge confidently to one of an infinite family of indistinguishable parameter vectors. This section asks whether the FSA-v2 model defined in Section 2 suffers from such a pathology, settles the matter analytically, and reports the empirical fingerprint of the diagnosis. The detailed derivations of the rank-deficient parameter pairs and of the G1 reparametrization are deferred to Appendix A.

3.1 Fisher information and local identifiability

For a parameter vector $\theta \in \mathbb{R}^d$ and an observation record y, the score function is $s(\theta; y) = \nabla_\theta \log p(y \mid \theta)$ and the *Fisher information matrix* (FIM) is

$$\mathcal{I}(\theta) \;=\; \mathbb{E}_{y \sim p(\cdot \mid \theta)}\big[\, s(\theta; y)\, s(\theta; y)^\top \big]. \tag{12}$$

$\mathcal{I}(\theta)$ is symmetric positive semidefinite, and two facts about it explain why it is the right diagnostic.

Proposition 2 (Local identifiability)**.** *Suppose $\theta \mapsto p(y \mid \theta)$ is twice differentiable in a neighbourhood of θ^*. Then θ^* is locally identifiable if and only if $\mathcal{I}(\theta^*)$ has full rank d.*

Proposition 3 (Cramér–Rao)**.** *Any unbiased estimator $\hat{\theta}$ of θ has covariance bounded below in the positive-semidefinite ordering by $\mathcal{I}(\theta)^{-1}$ (when $\mathcal{I}$ is invertible).*

In words: the FIM *rank* tells you whether the data can pin θ down at all, and the FIM *inverse* tells you how sharply. A rank-deficient FIM means there is a direction in parameter space along which the data are silent – two distinct θ vectors produce indistinguishable observation distributions – and any procedure trying to estimate θ in that direction will be at the mercy of the prior, not the data.

3.2 The pre-G1 model has three rank-deficient pairs

The four-channel observation model (8)–(11) probes the latent state (B, F, A) through linear-in-state likelihoods, and the latent state in turn carries information about the dynamics parameters via the drift (3)–(5). At a one-day window the slow Banister state moves only marginally, so the inferential signal about the dynamics parameters is concentrated in how $\partial(B, F, A)/\partial\theta$ couples through the observation likelihood.

A direct calculation along these lines (deferred to Appendix A) reveals that, near the operating point of Definition 1, three pairs of pre-G1 parameters enter the one-day likelihood through a single linear combination each, so that the contribution of each pair to the FIM is rank-one rather than rank-two. The three pairs and their identifiable combinations are summarised in Table 2.

The intuitive story behind each pair is the same: at one-day windows the system rarely strays far from the operating point, so any parameter whose effect manifests only via departures from that operating point is invisible to the data. For instance, in the pair $(\kappa_B, \varepsilon_A)$ the gain κ_B multiplies the daily training stimulus, while ε_A multiplies κ_B by a factor close to one whenever $A \approx A_{\text{typ}}$. Only the *product* $\kappa_B(1 + \varepsilon_A A_{\text{typ}})$ is felt by the data; the individual factors are not.

3.3 The G1 reparametrization absorbs the three pairs

The cure is a simple coordinate change: rotate each rank-deficient pair into a strongly identified linear combination and a residual that vanishes at the operating point. The G1 reparametrization, already shown in equations (2)–(5), performs this rotation. Table 2 lists the three combinations side by side.

Pair	Strongly identified	Residual	Vanishing condition
$(\kappa_B, \varepsilon_A)$	$\kappa_B^{\text{new}} = \kappa_B(1 + \varepsilon_A A_{\text{typ}})$	ε_A	$A = A_{\text{typ}}$
(μ_F, μ_{FF})	$\mu_F^{\text{new}} = \mu_F + 2F_{\text{typ}}\mu_{FF}$	μ_{FF}	$F = F_{\text{typ}}$
(τ_F, λ_A)	$\tau_F^{\text{new}} = \tau_F/(1 + \lambda_A A_{\text{typ}})$	λ_A	$A = A_{\text{typ}}$

Table 2: The Stage G1 reparametrization. Each rank-deficient parameter pair of the pre-G1 model is rotated into a strongly identified combination (left column) and a residual (middle column) that vanishes at the operating point of Definition 1. The parameter *names* are kept; only the meanings and truth values change. There is also an implicit shift $\mu_0 \mapsto \mu_0 + \mu_{FF}F_{\text{typ}}^2$ that absorbs the constant from completing the square in (2).

The reparametrized drift is mathematically equivalent to the pre-G1 drift under a fixed relabelling of parameters; the equivalence (stated as Proposition 15 in Appendix A) is verified numerically by a unit test in `tests/test_g1_reparam.py` that confirms the two drift formulas agree to floating-point precision over 50 random states.

3.4 Empirical evidence of identifiability after G1

The analytical argument predicts full FIM rank at the operating point. The implementation furnishes empirical confirmation at one-day rolling windows on simulated truth data.

The numbers in Table 3 confirm the analytical prediction. The G1 form is identifiable at one-day windows; the strongly identified parameters $(\kappa_B, \tau_F, \mu_F)$ retain their original prior widths; and the residuals $(\varepsilon_A, \lambda_A, \mu_{FF})$, whose posteriors were formerly artificially wide because of the rank deficiency, now contract to roughly a sixth of their pre-G1 spread under tightened priors that the rank analysis itself motivates.

Remark 4 (Where the FIM analysis lives in the code). The analytical FIM derivation summarised here – and developed in full in Appendix A – is recorded in the project's commit log (Stage G1)

Quantity	Pre-G1	Post-G1
rank$\widehat{\mathcal{I}}$ at one-day window	$d-3$	d
parameters covered by 90 % credible interval	27/30	**30/30**
identifiable estimable parameters	3/6	**6/6**
posterior std on ε_A (lognormal scale)	0.093	**0.016**
90 % CI for ε_A	[0.275, 0.550]	[**0.373**, **0.423**]

Table 3: Identifiability before and after the G1 reparametrization, measured at one-day rolling-window scale on simulated truth data. "Estimable" counts the six parameters that the FIM analysis identified as candidates for tightening. Numbers from `version_2/outputs/fsa_high_res/` and the Stage G1 commit record. The residual standard deviation on ε_A tightens by roughly a factor of six; analogous tightening is observed on the other two residuals μ_{FF} and λ_A.

rather than as standalone code. A useful follow-up would be to add a numerical FIM module under `smc2fc/diagnostics/fim/` that reproduces these results from the JAX dynamics by automatic differentiation, so that any future modification of the model triggers a re-run of the rank check automatically.

4 Stability of the FSA-v2 SDE: a Lyapunov analysis

Identifiability tells us that the inference problem for θ is well-posed; stability tells us that the dynamical system itself is well-behaved. This section asks two stability questions about the FSA-v2 SDE in its G1-reparametrized form: where are the equilibria of its deterministic skeleton, and is the full stochastic process ergodic on its physiological operating domain? The first question admits a clean answer in terms of the Stuart–Landau growth rate; the second is settled affirmatively via a quadratic Lyapunov function. Detailed Jacobian computations, the Foster–Lyapunov drift calculation, and bounds on constants are deferred to Appendix B.

4.1 Deterministic skeleton: equilibria and the Stuart–Landau bifurcation

Set $\sigma_B = \sigma_F = \sigma_A = 0$ in (3)–(5) and a constant control $\Phi(t) \equiv \Phi_c$. The deterministic skeleton has an explicit $A = 0$ branch

$$B_0^*(\Phi_c) = \frac{\kappa_B \tau_B \Phi_c}{1 + \varepsilon_A A_{\text{typ}}}, \quad F_0^*(\Phi_c) = \kappa_F \tau_F (1 + \lambda_A A_{\text{typ}})\, \Phi_c, \quad A_0^* = 0, \tag{13}$$

together with an implicit oscillatory branch where $A^2 = \mu(B, F)/\eta$. Local stability of the $A = 0$ branch is determined entirely by the sign of the Stuart–Landau growth rate at the equilibrium, as the following result records.

Proposition 5 (Local stability of the $A = 0$ branch)**.** *The deterministic equilibrium* (13) *is locally asymptotically stable if and only if* $\mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) < 0$.

The proof is short: the drift Jacobian at (13) turns out to be upper triangular, with two unconditionally negative eigenvalues coming from the slow Banister directions and a third eigenvalue equal to $\mu(B_0^*, F_0^*)$ controlling the autonomic amplitude. Details are in Appendix B.

The bifurcation locus $\{\Phi_c : \mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) = 0\}$ turns out to be a single positive root $\Phi_c^{\text{crit}} \approx 7.5$ TRIMP-units per day at the truth values of Table 1. The picture this paints is physiologically intuitive: at the canonical baseline $\Phi_c = 1$, $\mu(B_0^*, F_0^*) \approx 0.13 > 0$, so the $A = 0$ equilibrium is unstable and the model predicts that an athlete training at the default load settles on a non-trivial autonomic oscillation. Above the critical training load $\Phi_c \approx 7.5$, the quadratic overtraining-collapse term $\mu_{FF}(F - F_{\text{typ}})^2$ dominates, μ falls back through zero, and the system loses its oscillation: the autonomic amplitude collapses to zero, consistent with empirical observations that very large chronic training loads suppress autonomic variability.

4.2 Stochastic stability and exponential ergodicity

We turn to the full SDE (3)–(5) on the operating domain $\mathcal{X} = [0,1] \times [0,\infty) \times [0,\infty)$. The state-dependent diffusions vanish at the boundary $\partial\mathcal{X}$, so $\mathcal{X}$ is positively invariant. We seek a single quantitative statement that captures the long-time behaviour of the process. The Foster–Lyapunov framework supplies the right object.

Definition 6 (Foster–Lyapunov drift)**.** A function $V : \mathcal{X} \to [0,\infty)$ is said to satisfy a Foster–Lyapunov drift inequality with rate $\alpha > 0$ and constant $C \geq 0$ if the infinitesimal generator $\mathcal{L}$ of the SDE satisfies

$$\mathcal{L}V(x) \;\leq\; -\alpha\, V(x) + C \quad \text{for all } x \in \mathcal{X}. \tag{14}$$

Inequality (14) is the standard hypothesis that delivers exponential ergodicity in the sense of total-variation convergence to a unique invariant distribution (Khasminskii, 1980; Meyn and Tweedie, 1993). We exhibit a V that satisfies it.

Theorem 7 (Stochastic stability of FSA-v2)**.** *Under bounded controls $\Phi : \mathbb{R}_+ \to [0, \Phi_{\max}]$, the FSA-v2 SDE (3)–(5) satisfies the Foster–Lyapunov drift inequality (14) for the quadratic Lyapunov function*

$$V(B,F,A) \;=\; \tfrac{1}{2}\big(aB^2 + bF^2 + cA^2\big), \qquad a,b,c > 0, \tag{15}$$

with rate $\alpha = \min\big(2/\tau_B,\, 1/[\tau_F(1+\lambda_A A_{\text{typ}})]\big)$ and a finite constant C depending on the truth parameters, $\Phi_{\max}$, and the choice of a, b, c.

The proof is a routine but tedious computation of $\mathcal{L}V$ component by component, bounding each term using the boundedness of B on $[0,1]$ together with two applications of Young's inequality on the F and A components; we record it in Appendix B. The intuitive picture is that the three dissipative terms in the drift (linear decay of B, linear decay of F, cubic damping of A) overwhelm the linear growth and noise contributions outside a bounded region of state space, which is exactly what the Foster–Lyapunov inequality formalises.

Theorem 8 (Khasminskii, 1980; Meyn–Tweedie, 1993)**.** *Let $X_t = (B_t, F_t, A_t)$ be the FSA-v2 process under a fixed bounded control Φ, satisfying the conditions of Theorem 7. Then X admits a unique invariant probability distribution π^Φ on $\mathcal{X}$, and there exist constants $K_0 > 0$ and $\rho \in (0,\alpha)$ such that, for every $x_0 \in \mathcal{X}$,*

$$\big\|\mathrm{Law}(X_t \mid X_0 = x_0) - \pi^\Phi\big\|_{\mathrm{TV}} \;\leq\; K_0\,(1 + V(x_0))\, e^{-\rho t} \quad \textit{for all } t \geq 0, \tag{16}$$

where $\|\cdot\|_{\mathrm{TV}}$ is the total-variation norm.

The exponential rate ρ inherits the slowest dissipative timescale of the model. With $\tau_B = 42$ days and $\tau_F^{\text{old}} := \tau_F(1+\lambda_A A_{\text{typ}}) = 7$ days, the rate is $\alpha = 1/7\,\mathrm{d}^{-1}$ and convergence to invariance has half-life roughly $7\log 2 \approx 4.85$ days. This is consistent with the rolling-window choice of one-day windows and twelve-hour strides in Section 6: the SMC^2 filter operates on a fundamental dynamical scale comparable to the relaxation time of the process being filtered.

Remark 9 (Limitations of the bound)*.* The proof of Theorem 7 in Appendix B uses the simplest possible quadratic Lyapunov function and crude global bounds on the residual factors of the drift. The constant C that emerges is therefore loose: it scales with $\Phi_{\max}^2$ rather than with the time-average of Φ^2, and ignores the cancellations that the real dynamics exhibit near the operating point. A sharper analysis – perhaps with a piecewise Lyapunov function that switches between the regimes $\mu < 0$ and $\mu > 0$ identified in Proposition 5 – would deliver a tighter rate and a smaller constant. Such a refinement is left for future work.

5 The bootstrap particle filter

5.1 Bayesian filtering recursion

Let $X_n \in \mathbb{R}^3$ denote the latent state (B, F, A) at the nth fifteen-minute time step, and let Y_n denote the corresponding multi-channel observation. For a fixed value of the static parameter θ_{dyn}, the FSA-v2 model defines a hidden Markov model with transition density

$$X_n \mid X_{n-1} \sim p(\,\cdot \mid X_{n-1},\, \Phi_n,\, \theta_{\text{dyn}}) \tag{17}$$

implicitly through the Euler–Maruyama discretisation of (3)–(5), and observation density

$$Y_n \mid X_n \sim p(\,\cdot \mid X_n,\, C(t_n),\, \theta_{\text{dyn}}) \tag{18}$$

given by the four-channel model (8)–(11). Bayesian filtering is the recursive computation of the conditional distribution $\pi_n(\mathrm{d}x_n) := p(\mathrm{d}x_n \mid Y_{1:n},\, \theta_{\text{dyn}})$. It factorises into a one-step prediction and a Bayes update:

$$\pi_{n|n-1}(\mathrm{d}x_n) = \int p(\mathrm{d}x_n \mid x_{n-1},\, \Phi_n,\, \theta_{\text{dyn}})\, \pi_{n-1}(\mathrm{d}x_{n-1}), \tag{19}$$

$$\pi_n(\mathrm{d}x_n) \propto p(Y_n \mid x_n,\, C(t_n),\, \theta_{\text{dyn}})\; \pi_{n|n-1}(\mathrm{d}x_n). \tag{20}$$

Equations (19)–(20) have closed form only when both densities are linear-Gaussian (the Kalman case). For FSA-v2 they do not, and a Monte Carlo approximation is required.

5.2 Sequential importance resampling

The bootstrap particle filter (Gordon, Salmond and Smith, 1993; Doucet and Johansen, 2011) approximates π_n by a weighted empirical measure

$$\pi_n^N(\mathrm{d}x_n) \;=\; \sum_{i=1}^{N} W_n^{(i)}\, \delta_{X_n^{(i)}}(\mathrm{d}x_n), \qquad \sum_{i=1}^{N} W_n^{(i)} = 1, \tag{21}$$

on a population of N *state particles* $\{X_n^{(i)}\}$. The recursion has three substeps: (i) propagate each particle through the dynamics, (ii) reweight by the observation likelihood, (iii) resample to prevent weight degeneracy. This is sequential importance resampling (SIR), summarised in Algorithm 1. Resampling is performed *systematically*: with $u_0 \sim \mathcal{U}[0, 1)$, the resampled indices are obtained by walking the cumulative weight $C(j) = \sum_{i \le j} W_n^{(i)}$ at the offsets $u_k = (k + u_0)/N$ for $k = 0, \ldots, N-1$. Systematic resampling is $O(N)$, has lower Monte Carlo variance than multinomial resampling, and is deterministic given u_0.

Algorithm 1 Bootstrap SIR for the FSA-v2 filter, given θ_{dyn}

1: Sample $X_0^{(i)} \sim p(\mathrm{d}x_0 \mid \theta_{\text{dyn}})$ for $i = 1, \ldots, N$; set $W_0^{(i)} = 1/N$.
2: **for** $n = 1, 2, \ldots$ **do**
3: **Propagate.** For each i, draw $X_n^{(i)} \sim p(\,\cdot \mid X_{n-1}^{(i)},\, \Phi_n,\, \theta_{\text{dyn}})$ by one Euler–Maruyama step of (3)–(5).
4: **Reweight.** Set $\tilde{W}_n^{(i)} = W_{n-1}^{(i)}\, p(Y_n \mid X_n^{(i)},\, C(t_n),\, \theta_{\text{dyn}})$ and $W_n^{(i)} = \tilde{W}_n^{(i)} / \sum_j \tilde{W}_n^{(j)}$.
5: **ESS check.** Compute $N_{\text{eff}} = 1/\sum_i (W_n^{(i)})^2$. If $N_{\text{eff}} < N_{\text{thr}}$, resample $\{X_n^{(i)}\}$ systematically by $\{W_n^{(i)}\}$ and reset $W_n^{(i)} \leftarrow 1/N$.
6: **end for**

The implementation in `smc2fc/filtering/gk_dpf_v3_lite.py` adds two refinements that we mention but do not develop here. First, when parameter particles are coupled to the state

particles in the SMC^2 outer layer of Section 6, a Liu–West shrinkage move is applied with an effective-sample-size dependent bandwidth, slowing parameter-cloud collapse. Second, when the effective sample size collapses despite resampling, an optional low-rank Sinkhorn optimal transport step blends a transport plan into the systematic resample. Both are diagnostic / numerical safeguards rather than changes to the mathematical statement of Algorithm 1.

5.3 The unbiased marginal likelihood

A key property of the bootstrap filter, which makes the SMC^2 algorithm of the next section possible, is that it produces an unbiased estimator of the marginal likelihood. With the unnormalised weights $\tilde{W}_n^{(i)}$ from line 4 of Algorithm 1, define

$$\widehat{L}_N(\theta_{\text{dyn}}) \;:=\; \prod_{n=1}^{T}\Big(\frac{1}{N}\sum_{i=1}^{N}\tilde{W}_n^{(i)}\Big). \tag{22}$$

Proposition 10 (Unbiasedness; Del Moral, 2004)**.** *For any $N \geq 1$ and any $T \geq 1$, $\mathbb{E}[\,\widehat{L}_N(\theta_{\text{dyn}})\,] = p(Y_{1:T} \mid \theta_{\text{dyn}})$.*

A proof is in Del Moral (2004, Thm. 7.4.2). For the present notes the significance of Proposition 10 is operational: it allows us to plug $\widehat{L}_N(\theta_{\text{dyn}})$ in place of the (intractable) exact likelihood $p(Y_{1:T} \mid \theta_{\text{dyn}})$ inside any sampler over θ_{dyn} and obtain a procedure that targets exactly the right posterior. This is precisely what SMC^2 does.

6 SMC^2 over parameters

6.1 The tempered posterior over θ_{dyn}

The bootstrap filter of Section 5 assumes the static parameter θ_{dyn} is known. In practice θ_{dyn} is itself the principal object of inference, with a prior $p(\theta_{\text{dyn}})$ and posterior

$$p(\theta_{\text{dyn}} \mid Y_{1:T}) \;\propto\; p(\theta_{\text{dyn}})\, p(Y_{1:T} \mid \theta_{\text{dyn}}). \tag{23}$$

A direct sampler is unavailable because $p(Y_{1:T} \mid \theta_{\text{dyn}})$ has no closed form. The SMC^2 algorithm of Chopin, Jacob and Papaspiliopoulos (2013) replaces this likelihood by the unbiased estimator $\widehat{L}_N(\theta_{\text{dyn}})$ from (22) and, for a fixed schedule $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_K = 1$, defines a sequence of tempered targets

$$\pi_\lambda(\theta_{\text{dyn}}) \;\propto\; p(\theta_{\text{dyn}})^{1-\lambda}\,\widehat{L}_N(\theta_{\text{dyn}})^{\lambda}, \qquad \lambda \in [0,1]. \tag{24}$$

At $\lambda = 0$ the target is the prior; at $\lambda = 1$ it is, in expectation, the posterior. The interpolation in between bridges the two in a way that keeps the importance weights well-behaved.

6.2 Adaptive tempering and the move step

A population of M *parameter particles* $\{\theta_{\text{dyn}}^{(m)}\}_{m=1,\dots,M}$ is propagated through the schedule $\{\lambda_k\}$. At each step the unnormalised tempering weight

$$\omega_k^{(m)} \;=\; \Big(\frac{\widehat{L}_N(\theta_{\text{dyn}}^{(m)})}{p(\theta_{\text{dyn}}^{(m)})^{-1}}\Big)^{\lambda_k-\lambda_{k-1}} \tag{25}$$

is computed, and an effective sample size $M_{\text{eff}} = (\sum_m \omega_k^{(m)})^2 / \sum_m (\omega_k^{(m)})^2$ is monitored. The interval $\lambda_k - \lambda_{k-1}$ is chosen *adaptively*, by bisection, to drive M_{eff}/M to a prescribed target (typically 0.5); this avoids both unnecessarily short bridges and weight collapse. After resampling, each parameter particle is moved by a Hamiltonian Monte Carlo (HMC) kernel that targets π_{λ_k},

with the mass matrix re-estimated from the current particle cloud. Algorithm 2 summarises the procedure.

Algorithm 2 Adaptive-tempered SMC^2 over θ_{dyn}

1: Sample $\theta_{\text{dyn}}^{(m)} \sim p(\theta_{\text{dyn}})$ for $m = 1, \ldots, M$; set $\lambda_0 = 0$.
2: For each m, run Algorithm 1 with $\theta_{\text{dyn}}^{(m)}$ to obtain $\widehat{L}_N(\theta_{\text{dyn}}^{(m)})$.
3: **while** $\lambda_k < 1$ **do**
4: Solve for the smallest increment $\delta \in (0, 1 - \lambda_k]$ such that the tempering ESS ratio with weights $\omega^{(m)} \propto \widehat{L}_N(\theta_{\text{dyn}}^{(m)})^\delta$ equals the target (e.g. 0.5). Set $\lambda_{k+1} = \lambda_k + \delta$.
5: Resample $\{\theta_{\text{dyn}}^{(m)}\}$ systematically by $\{\omega^{(m)}\}$.
6: **Move.** Run a small number of HMC steps targeting $\pi_{\lambda_{k+1}}$ on each $\theta_{\text{dyn}}^{(m)}$, re-running Algorithm 1 as needed to evaluate the unbiased likelihood at proposed values.
7: $k \leftarrow k + 1$.
8: **end while**

6.3 Rolling-window SMC^2 and the Schrödinger–Föllmer bridge

When data arrive sequentially, running Algorithm 2 from scratch over the full record at every new arrival is wasteful. The implementation in this code base instead operates over a sliding sequence of *windows*: with a fifteen-minute time grid, each window is one day long (96 bins), the stride between consecutive windows is twelve hours (48 bins), and the macrocycle of fourteen days yields $(14 \cdot 96 - 96)/48 + 1 = 27$ windows.

For window $w \geq 2$ the parameter cloud is *warm-started* from the posterior of window $w - 1$. The transition is mediated by a Schrödinger–Föllmer bridge between two Gaussian fits, parameterised along the Bures–Wasserstein geodesic in the manifold of positive-definite covariance matrices. Concretely, let (m_0, Σ_0) denote a Gaussian moment match to the previous window's posterior, and let (m_1, Σ_1) denote an importance-weighted moment match obtained by a single particle-filter evaluation per particle on the new window's data. For a blend parameter $\rho \in [0, 1]$ the bridge measure has mean and covariance

$$m_\rho = (1 - \rho)\, m_0 + \rho\, m_1, \tag{26}$$

$$\Sigma_\rho = \big((1 - \rho)I + \rho\, T\big)\, \Sigma_0\, \big((1 - \rho)I + \rho\, T\big)^\top, \tag{27}$$

where $T = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2}$ is the Bures–Wasserstein optimal-transport map. The default implementation uses $\rho = 0.7$, biasing the warm-start toward the new window's evidence. We do not develop the geometry here; the reader unfamiliar with it can treat the bridge as a black box that produces a sensible Gaussian initialisation for the next window's tempering schedule.

7 Control as inference

7.1 The cost functional

Sections 5 and 6 treat the dynamics parameters θ_{dyn} and the latent state X as the things to be inferred from a fixed schedule Φ. We now reverse the perspective: the dynamics parameters and the initial state are taken as given, and the schedule itself becomes the unknown.

For a fixed truth θ_{dyn}, an initial state x_0, and a candidate schedule $\Phi : [0, T] \to [0, \Phi_{\max}]$, define the trajectory cost

$$J(\Phi) \;=\; \mathbb{E}_{X|\Phi}\left[-\int_0^T A(t)\,\mathrm{d}t \;+\; \lambda_F \int_0^T \max\!\big(F(t) - F_{\max},\, 0\big)^2\,\mathrm{d}t \right], \tag{28}$$

where the expectation is over the law of $X = (B, F, A)$ under the SDE (3)–(5) driven by Φ. The first term rewards mean autonomic amplitude; the second is a soft barrier penalising trajectories that exceed the fatigue ceiling $F_{\max} = 0.40$. The barrier weight is $\lambda_F = 1$ in the reference implementation; no Φ^2 control penalty is included.

7.2 Schedule decoder

The schedule Φ is parameterised in finite dimensions by a vector $\theta_{\text{ctrl}} \in \mathbb{R}^8$ of Gaussian-RBF anchor coefficients. With a fixed basis $\{\varphi_a(t)\}_{a=1}^{8}$ (eight evenly spaced Gaussians on $[0, T]$ at the discretisation grid) and a logit bias $c_\Phi := \log\big(\Phi_{\text{def}}/(\Phi_{\max} - \Phi_{\text{def}})\big)$ chosen so that $\theta_{\text{ctrl}} = 0$ produces the canonical-Banister default $\Phi \equiv \Phi_{\text{def}} = 1.0$, the decoder is

$$\Phi(t;\, \theta_{\text{ctrl}}) \;=\; \Phi_{\max} \cdot \sigma\!\left(c_\Phi + \sum_{a=1}^{8} (\theta_{\text{ctrl}})_a\, \varphi_a(t) \right). \tag{29}$$

The sigmoid envelope keeps $\Phi(t;\, \theta_{\text{ctrl}}) \in [0, \Phi_{\max}]$ for every θ_{ctrl}; the search is unconstrained in $\mathbb{R}^8$.

7.3 Tempered control posterior

Following the control-as-inference tradition (Kappen, 2005; Toussaint, 2009; Levine, 2018), we define a probability measure on θ_{ctrl} by treating the negative cost as an unnormalised log-likelihood. With inverse temperature $\beta > 0$ and a Gaussian prior $\theta_{\text{ctrl}} \sim \mathcal{N}(0,\, \sigma^2_{\text{prior}} I_8)$, the *control posterior* is

$$q_\beta(\theta_{\text{ctrl}}) \;\propto\; \exp\!\big(-\beta\, J(\theta_{\text{ctrl}})\big)\, \mathcal{N}(\theta_{\text{ctrl}};\, 0,\, \sigma^2_{\text{prior}} I_8), \tag{30}$$

where we now write $J(\theta_{\text{ctrl}}) := J(\Phi(\,\cdot\,; \theta_{\text{ctrl}}))$. Sampling from q_β is performed by exactly the SMC^2 machinery of Section 6, but with the unbiased filter likelihood $\widehat{L}_N(\theta_{\text{dyn}})$ replaced by the cost-likelihood $\exp(-\beta\, J(\theta_{\text{ctrl}}))$. The inverse temperature β is calibrated automatically from the cost spread of the prior cloud, and the tempered schedule $\lambda \in [0, 1]$ smoothly interpolates from prior to control posterior.

7.4 Posterior mean as the controller output

Proposition 11 (Posterior mean as a regularised cost minimiser)**.** *Let q_β be the control posterior* (30) *and write $\bar{\theta} := \mathbb{E}_{q_\beta}[\theta_{\text{ctrl}}]$. Then*

$$\bar{\theta} \;=\; \arg\min_{\theta \in \mathbb{R}^8} \Big\{ \mathbb{E}_{q_\beta}\big[\, \|\theta - \theta_{\text{ctrl}}\|_2^2 \,\big] \Big\}, \tag{31}$$

i.e. the posterior mean is the L^2 minimiser of squared distance under the tempered measure. In particular, when J is approximately quadratic near its minimum and the prior variance is large enough that the prior contribution to $\nabla \log q_\beta$ is small, $\bar{\theta}$ sits in a neighbourhood of the unregularised minimiser $\arg\min_\theta J(\theta)$ *of size $O(\beta^{-1/2})$.*

The implementation uses $\bar{\theta}$, computed as the equally-weighted mean of the SMC^2 parameter cloud at $\lambda = 1$, as the controller output. The corresponding schedule is then $\Phi(\,\cdot\,; \bar{\theta})$ via (29). We will see in Section 9 that this point output is sufficient in practice, despite throwing away the higher-order moments of q_β; Section 8.7 discusses why.

8 Receding-horizon model-predictive control

This section is the conceptual core of the document. It develops the finite-horizon stochastic optimal control problem, the receding-horizon principle, the various stochastic and output-feedback variants, and then specialises everything to the FSA-v2 closed-loop model-predictive

controller in both mathematical and implementation form. A closing subsection sketches a linear–quadratic–Gaussian approximation that could be added as a cheap baseline.

8.1 Finite-horizon stochastic optimal control

Consider a discrete-time stochastic system on a state $X_k \in \mathcal{X}$ and a control $U_k \in \mathcal{U}$, with transition kernel $X_{k+1} \sim p(\cdot \mid X_k, U_k)$, a stage cost $\ell : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$, and a terminal cost $V_f : \mathcal{X} \to \mathbb{R}$. The *open-loop* finite-horizon problem of length H is

$$\min_{u_{0:H-1} \in \mathcal{U}^H} \mathbb{E}\left[\sum_{k=0}^{H-1} \ell(X_k, u_k) + V_f(X_H) \,\middle|\, X_0 = x \right], \tag{32}$$

with the expectation taken over the joint law of $X_{1:H}$ given the chosen sequence $u_{0:H-1}$. Denote its optimal value by $V_H(x)$ and a minimising sequence by $u^*_{0:H-1}(x)$. The mapping $x \mapsto u^*_0(x)$ is the open-loop optimal first-action policy.

For FSA-v2 the state $X_k = (B_k, F_k, A_k)$ lives in $\mathcal{X} = [0,1] \times [0,\infty)^2$, and the "control sequence" is the eight-dimensional RBF-coefficient vector θ_{ctrl} that parameterises the entire schedule $\Phi(\,\cdot\,; \theta_{\text{ctrl}})$ over $[0,T]$ via (29). The discrete-time abstract notation (32) therefore maps onto the continuous schedule of Section 7 by a one-to-one identification between admissible $u_{0:H-1}$ and admissible θ_{ctrl}. The expectation in (32) is the same one already written out in (28).

8.2 The receding-horizon principle

In *receding-horizon* or *model-predictive* control, the open-loop problem (32) is solved repeatedly on a sliding window. At each replan time t_n, given a current state estimate $\widehat{x}_n$, one

1. solves (32) for $u^*_{0:H-1}(\widehat{x}_n)$;
2. applies only the first action $u^*_0(\widehat{x}_n)$ over a stride of length $\Delta < H$;
3. observes new data, updates the state estimate to $\widehat{x}_{n+1}$, and returns to step 1.

The resulting closed-loop policy is a function of $\widehat{x}_n$ alone, so the controller is implementable. Stability of the closed loop is delicate. Mayne and Rawlings (2000) show that under suitable assumptions on the horizon length and the terminal cost / terminal set, the closed-loop state remains in a positively invariant set and converges to a neighbourhood of a desired equilibrium. For FSA-v2 we rely on the *long-horizon* variant of that theory: the planning horizon H is chosen long enough relative to the slowest time constant $\tau_B = 42$ days that the truncation error from neglecting trajectory beyond H is small.

8.3 Stochastic and output-feedback variants

Three flavours of stochastic MPC are commonly distinguished.

(i) *Certainty-equivalent* MPC. Replace the expectation in (32) by evaluation at a point estimate $\hat{X} = \mathbb{E}[X \mid \widehat{x}_n]$, and solve the resulting deterministic problem. Cheap, but discards uncertainty information.

(ii) *Expected-cost* (stochastic) MPC. Keep the expectation, and approximate it by Monte Carlo (sampling trajectories of the stochastic dynamics) or by scenario trees.

(iii) *Output-feedback* MPC. When X_k is not observed directly, the controller acts on a state estimate $\widehat{x}_n$ produced by an external estimator (Kalman, particle filter, SMC^2, etc.). For nonlinear non-Gaussian systems no *separation principle* holds in general — the joint design of the estimator and the controller is part of the modelling, not a free composition.

The FSA-v2 pipeline is in class (ii) and (iii) simultaneously. The state is unobserved (only the four-channel proxies of (8)–(11) are available), so a state estimator is required: this is the rolling-window SMC^2 filter of Section 6. The expectation in the cost (28) is kept, evaluated by Monte Carlo on the SDE under common random numbers, and minimised in θ_{ctrl}-space by the control SMC^2 of Section 7. The full closed-loop policy is therefore the composition

$$(\text{observations}) \xrightarrow{\text{rolling SMC}^2\text{ filter}} \widehat{x}_n \xrightarrow{\text{control SMC}^2\text{, posterior mean}} \bar{\theta} \xrightarrow{(29)} \Phi(\,\cdot\,;\bar{\theta}) \xrightarrow{\text{plant}} (\text{new observations}).$$

There is no separation principle here. The fact that this composition works at all is an empirical claim, validated in Section 9.

8.4 The FSA-v2 MPC pipeline — mathematical statement

We now write down the closed-loop algorithm. Let $\Delta_w = 96$ bins be the filter window length (one day), $\Delta_s = 48$ bins the stride (twelve hours), and $H_T = 14 \cdot 96 = 1344$ bins the macrocycle (fourteen days), so that the number of windows is $N_w = (H_T - \Delta_w)/\Delta_s + 1 = 27$.

Algorithm 3 FSA-v2 closed-loop MPC

1: Initialise the plant at (B_0, F_0, A_0); initialise an empty observation buffer $Y_{1:0} = \emptyset$.
2: **for** $n = 1, \ldots, N_w$ **do**
3: **Filter update.** If $n = 1$, run cold-start SMC^2 (Algorithm 2) on $Y_{(n-1)\Delta_s+1:(n-1)\Delta_s+\Delta_w}$ to produce a posterior over θ_{dyn} for the first window; otherwise warm-start from the previous window's posterior via the Schrödinger–Föllmer bridge of Section 6.
4: **State extraction.** For a small set of θ_{dyn} particles drawn from the posterior, run the bootstrap filter (Algorithm 1) forward to the stride endpoint and compute the importance-weighted mean of the resulting (B, F, A); call this $\widehat{x}_n$.
5: **Control plan.** Run the control SMC^2 of Section 7, conditioned on $\widehat{x}_n$ and the posterior-mean dynamics parameters $\bar{\theta}_{\text{dyn}}$, to obtain the posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$.
6: **Apply.** Decode $\Phi(\,\cdot\,;\bar{\theta}_{\text{ctrl}})$ via (29) and apply only its first stride (length Δ_s) to the plant; collect the resulting observations and append them to the buffer.
7: **end for**

The horizon over which the controller plans (line 5) is the full macrocycle remainder $H_T - (n-1)\Delta_s$, but only the first Δ_s bins of the plan are committed (line 6). This is the receding-horizon truncation of Section 8.2 applied to the FSA-v2 problem. Replanning occurs every twelve hours, at the wake-window boundary so that the morning-loaded burst envelope of (7) is delivered in full each day.

8.5 Code-level structure of the MPC implementation

This subsection departs from the usual mathematical idiom and describes how the abstract loop of Section 8.4 is realised in the source code of the project. The reader who only wants the mathematics may skip to Section 8.6; the reader who intends to run, modify, or extend the controller will find this useful. We name three artefacts.

(a) The plant interface, `StepwisePlant`. The "plant" in stochastic optimal control is whatever object the controller acts upon to produce new observations. In the code base it is the class `StepwisePlant` in `version_2/models/fsa_high_res/_plant.py`. A `StepwisePlant` carries the hidden true state (B, F, A), the true dynamics parameters, and a global bin counter; its single mutating method is

```
advance(stride_bins, Phi_daily) → (trajectory, obs_HR, obs_sleep, obs_stress,
obs_steps).
```

A call to `advance` expands the supplied daily-Φ array into the morning-loaded sub-daily envelope of (7) (preserving $\int_{\text{day}} \Phi \, \mathrm{d}t = 24\,\Phi_d$ exactly), runs Euler–Maruyama for `stride_bins` fifteen-minute steps, samples the four-channel observations of (8)–(11), and updates the internal state. Pedagogically, the `StepwisePlant` is the digital embodiment of the "real world" against which the controller is evaluated; mathematically, a single call samples one transition of the joint state-observation process under the chosen control over a stride of length Δ_s.

(b) The MPC driver, `bench_smc_full_mpc_fsa.py`. The receding-horizon orchestrator lives in `version_2/tools/bench_smc_full_mpc_fsa.py`. Its execution sequence implements Algorithm 3 step by step. For each window n it

1. runs the rolling-window filter by calling `run_smc_window` (cold-start, $n = 1$) or `run_smc_window_bridge` with a Schrödinger–Föllmer base measure (warm-start, $n \geq 2$);
2. extracts the posterior-mean dynamics parameters and the smoothed latent state $\widehat{x}_n$ at the end of the window, the latter computed by running Algorithm 1 forward on a small set of θ_{dyn} particles drawn from the posterior and taking the weighted mean of the resulting (B, F, A) samples;
3. assembles a `ControlSpec` (cost functional, horizon length, current state estimate, RBF basis);
4. calls the control SMC^2 loop in `smc2fc/control/tempered_smc_loop.py`, which returns a posterior-mean schedule coefficient $\bar{\theta}_{\text{ctrl}}$ and a corresponding Φ schedule;
5. calls `plant.advance(stride_bins, Phi_daily)` to apply only the first stride of that schedule;
6. appends the resulting observations to the data buffer and shifts the window.

(c) The posterior-mean handoff between windows. The smoothed state $\widehat{x}_n$ extracted in step 2 above is what initialises the next window's particle filter. Consecutive windows are therefore connected *at the state level* by a single point estimate (the importance-weighted smoothed mean) and *at the parameter level* by the SF Bures–Wasserstein bridge of Section 6. This is the practical realisation of the warm-start that gives the rolling SMC^2 its computational tractability: at fourteen days of fifteen-minute data the cold-start cost would be 27× larger.

Table 4 collects the mapping between the steps of Algorithm 3 and the source files that realise them.

Step in Algorithm 3	Concept	Source
3	rolling SMC^2 filter	`smc2fc/core/tempered_smc.py`
3	SF Bures–Wasserstein bridge	`smc2fc/core/sf_bridge.py`
3	inner bootstrap filter	`smc2fc/filtering/gk_dpf_v3_lite.py`
4	smoothed-state extraction	`version_2/tools/bench_smc_full_mpc_fsa.py`
5	control SMC^2 outer loop	`smc2fc/control/tempered_smc_loop.py`
5	cost functional, RBF decoder	`version_1/models/fsa_high_res/control.py`
6	plant advance	`version_2/models/fsa_high_res/_plant.py`
2,6	sub-daily Φ envelope	`version_2/models/fsa_high_res/_phi_burst.py`

Table 4: Mapping from Algorithm 3 to the source code.

8.6 Recursive feasibility and the F-barrier

The fatigue ceiling $F_{\max} = 0.40$ is enforced through the soft-barrier penalty $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ in the cost (28), not through a hard constraint. The choice is deliberate. Under stochastic dynamics with state-dependent diffusion and unbounded driving noise, an exact hard constraint $F(t) \leq F_{\max}$ cannot be guaranteed by any controller without making the admissible-control set empty for some achievable initial conditions; this is the standard *recursive feasibility* difficulty in stochastic MPC. A soft penalty preserves recursive feasibility – there is always a feasible schedule – and produces the constraint approximately, with the violation rate decreasing as λ_F is increased. Empirically (see Section 9) the closed-loop F-violation rate is 1.28% on the 84-day open-loop benchmark and 0.00% on the 14-day closed-loop benchmark, well within the 5% acceptance gate.

8.7 Robustness and the flat-cost-surface argument

The controller of Section 8.4 commits the posterior mean $\bar{\theta}_{\text{ctrl}}$ rather than a sample from the control posterior or the full distribution. One might worry that this discards useful information, especially when the filter posterior over θ_{dyn} is uncertain or biased. Empirically the policy is nevertheless robust to filter parameter drift, and the explanation is geometric: near the optimum, the cost $J(\theta_{\text{ctrl}})$ is locally flat in the directions explored by the SMC^2 cloud. A first-order Taylor expansion of J at the posterior mean gives, for a perturbed estimate $\bar{\theta} + \delta$,

$$J(\bar{\theta} + \delta) \;\approx\; J(\bar{\theta}) + \tfrac{1}{2}\delta^\top H \delta, \qquad H = \nabla^2 J(\bar{\theta}), \tag{33}$$

so the additional cost from using $\bar{\theta}$ in place of an unknown "oracle" optimum scales with the Hessian. When H is small in the directions of greatest filter uncertainty, the closed-loop performance is correspondingly insensitive to filter accuracy. Section 9 reports closed-loop performance on a record where only 3 out of 26 estimable parameters pass a strict coverage gate; the controller nevertheless achieves a +17% amplitude gain. The flat-cost-surface argument is the mechanism.

8.8 An exact-approximate alternative: LQG and the differential Riccati equation

We close Section 8 by sketching a second controller that could be added to the project as a cheap baseline or as a warm-start for the SMC^2 control of Section 7. The sketch is deliberately complete enough to make the assumptions and limitations explicit; it is not yet implemented in the code base, and is flagged as required next work in Section 10.3.

Linearisation around an operating point. Three assumptions reduce the FSA-v2 problem to one with a closed-form optimal feedback law.

Assumption 12. There is a nominal trajectory $(x^*(t), \Phi^*(t))$ on $[0, T]$, taken in the simplest case to be the deterministic-skeleton equilibrium $(B^*, F^*, A^*)(\Phi_{\text{const}})$ under a constant control. With $\tilde{x} = x - x^*$ and $\tilde{\Phi} = \Phi - \Phi^*$, define the linearised drift coefficients

$$A_{\text{lin}}(t) = \left.\frac{\partial f}{\partial x}\right|_*, \qquad B_{\text{lin}}(t) = \left.\frac{\partial f}{\partial \Phi}\right|_*, \tag{34}$$

where f is the drift of (3)–(5).

The cubic Stuart–Landau term $-\eta A^3$ is linearised away under Assumption 12, so the bistable / limit-cycle behaviour of the autonomic amplitude is lost. The linearisation is valid only in a neighbourhood of the operating point.

Assumption 13. The state-dependent Jacobi and CIR diffusions in (3)–(5) are replaced by constant-coefficient Gaussian noise $\bar{\sigma}\,\mathrm{d}W$ evaluated at the operating point.

The $[0,1]$ and $[0,\infty)$ state constraints are no longer enforced implicitly by the noise vanishing at the boundary, so trajectories may leave the physiological domain. Acceptable when fluctuations around the operating point are small relative to the boundary distances.

Assumption 14. The cost (28) is replaced by the quadratic surrogate

$$J_{\mathrm{LQ}}(\tilde{\Phi}) \;=\; \mathbb{E}\left[\int_0^T (\tilde{x}^\top Q\tilde{x} + \tilde{\Phi}^\top R\tilde{\Phi})\,\mathrm{d}t + \tilde{x}(T)^\top Q_T\tilde{x}(T)\right], \tag{35}$$

with $Q \succeq 0$, $R \succ 0$, $Q_T \succeq 0$.

To approximate the $-\int A\,\mathrm{d}t$ reward, one chooses $x_{\mathrm{ref}} \neq x^*$ shifted toward higher A and penalises $\tilde{x} = x - x_{\mathrm{ref}}$ with Q giving most weight to the A component; this preserves $Q \succeq 0$. The soft fatigue barrier $\lambda_F \cdot \max(F - F_{\max}, 0)^2$ is replaced by a quadratic in $F - F^*$, which penalises both directions symmetrically – too restrictive on the rest side and too lax on the over-fatigue side near the barrier.

The differential Riccati equation. Under Assumptions 12–14, the optimal feedback law for the linearised system is the time-varying linear policy

$$\tilde{\Phi}^*(t) \;=\; -K(t)\,\tilde{x}(t), \qquad K(t) \;=\; R^{-1}\,B_{\mathrm{lin}}(t)^\top\,P(t), \tag{36}$$

where $P : [0,T] \to \mathbb{R}^{3\times 3}$, $P(t) \succeq 0$, solves the *differential Riccati equation*

$$-\dot{P}(t) \;=\; A_{\mathrm{lin}}(t)^\top P(t) + P(t)\,A_{\mathrm{lin}}(t) - P(t)\,B_{\mathrm{lin}}(t)\,R^{-1}\,B_{\mathrm{lin}}(t)^\top P(t) + Q, \qquad P(T) = Q_T. \tag{37}$$

For a time-invariant linearisation (constant A_{lin}, B_{lin}) and an infinite horizon, P solves the *algebraic* Riccati equation $0 = A_{\mathrm{lin}}^\top P + P\,A_{\mathrm{lin}} - P\,B_{\mathrm{lin}}\,R^{-1}\,B_{\mathrm{lin}}^\top P + Q$. Under Assumptions 12–13 the state-estimation problem also linearises and is solved by the Kalman–Bucy filter; the LQG controller is then the certainty-equivalent composition of Kalman–Bucy filter and LQR feedback, and the (linear-Gaussian) separation principle holds.

Where this would fit in the project. A complete LQG companion to the SMC^2 controller would consist of a new module `smc2fc/control/lqg/` containing

1. automatic linearisation of the drift in `version_2/models/fsa_high_res/_dynamics.py` around a user-supplied operating point, via JAX automatic differentiation;
2. a numerical solver for (37) by backward integration on the same fifteen-minute time grid used elsewhere in the project;
3. a `LQGController` class emitting the feedback gain $K(t)$ of (36) and a corresponding Φ schedule on the project's time grid.

The natural uses are (i) as a *cheap baseline* against which the SMC^2 controller of Section 7 is benchmarked — does the non-linear posterior-mean controller actually outperform the linear LQG approximation, and by how much? — and (ii) as a *warm-start* for the control posterior, by least-squares projection of the LQG schedule onto the RBF basis to give an initial θ_{ctrl} around which the SMC^2 then explores. None of this currently exists in the repository; we return to it in Section 10.3.

9 Results

The two headline empirical results of the project are an open-loop benchmark on the canonical 84-day horizon (one chronic time constant of 42 days, doubled) and the closed-loop fourteen-day rolling controller. Both compare the SMC^2 schedule to the canonical constant baseline $\Phi \equiv 1.0$. Headline numbers are summarised in Table 5.

Benchmark	Horizon	$\bar{A}$ (SMC^2)	$\bar{A}$ (baseline)	gain	F-violation
Open-loop, Stage D	84 d	0.645	0.503	+28%	1.28%
Closed-loop MPC, Stage E5	14 d	0.0964	0.0823	+17%	0.00%

Table 5: Headline empirical results. $\bar{A}$ denotes the mean autonomic amplitude $\frac{1}{T}\int_0^T A(t)\,\mathrm{d}t$ averaged over Monte Carlo trajectories. Sources: `version_1/outputs/fsa_high_res/RESULT.md` (Stage D), `version_2/outputs/fsa_high_res/RESULT.md` (Stage E5).

9.1 Stage D: open-loop schedule on the 84-day horizon

In the open-loop benchmark, the dynamics parameters θ_{dyn} are treated as known at their truth values, the initial state is fixed at $(B_0, F_0, A_0) = (0.05, 0.30, 0.10)$ (a de-trained athlete with high residual fatigue), and the SMC^2 control of Section 7 is run *once* over the full 84-day horizon. The output is a single posterior-mean schedule $\Phi(\,\cdot\,;\bar{\theta})$.

The optimised schedule has mean $\Phi \approx 1.96$ over the horizon and a front-loaded periodisation profile (heavy training for the first ~ 60 days, taper thereafter). Mean amplitude $\bar{A} = 0.645$ exceeds the constant baseline $\bar{A} = 0.503$ by 28%, exceeds the sedentary reference $\bar{A} = 0.209$ by a factor of 3.08, and the soft fatigue barrier is violated only 1.28% of the time — well within the 5% acceptance gate of Section 8.6. All four acceptance gates (mean-A matches baseline within 3%, mean-A $\geq 1.40\times$ sedentary, mean-$\Phi \in [0.5, 2.5]$, F-violation $\leq 5\%$) pass.

9.2 Stage E5: closed-loop MPC on the 14-day macrocycle

In the closed-loop benchmark the dynamics parameters are no longer known to the controller. Algorithm 3 is run on a 14-day record of four-channel observations, with a one-day filter window, twelve-hour stride, and 27 replans. The plant is the `StepwisePlant` of Section 8.5.

Mean amplitude under the closed-loop policy is $\bar{A} = 0.0964$, against $\bar{A} = 0.0823$ for the constant baseline $\Phi \equiv 1.0$ on the same 14-day horizon. (The shorter horizon explains the smaller absolute numbers compared to Stage D: the integral $\int_0^T A(t)\,\mathrm{d}t$ has not yet converged to the asymptotic mean.) The +17% gain is achieved with a remarkable schedule shape: the controller commits a *rest-leaning* Φ in the range 0.25–0.31 across all 27 strides, against the canonical baseline of 1.0. Physiologically, this is the controller exploiting the fact that the simulated athlete starts with high residual fatigue ($F_0 = 0.30$): the optimal short-horizon strategy is to clear fatigue rather than to add training stimulus, which unlocks the Stuart–Landau growth-rate $\mu(B, F)$ and lets A rise from 0.10 to ~ 0.20 over the macrocycle.

The fatigue ceiling is never violated (0.00%). Wall-clock cost on a single GPU is 41 minutes for the full closed loop. The closed-loop policy is robust despite imperfect filter coverage — on this record only 3 of 26 estimable parameters meet a strict $\geq 5/6$ coverage gate at the window level — consistent with the flat-cost-surface argument of Section 8.7.

A trajectory plot of $A(t)$ under both policies, and a plot of the SMC^2 posterior-mean schedule, are included as Figure placeholders here; they can be regenerated from the artefacts in `version_2/outputs/fsa_high_res/` by re-running `tools/bench_smc_full_mpc_fsa.py`.

Figure placeholder: $A(t)$ trajectory under closed-loop MPC (blue) and constant baseline $\Phi \equiv 1$ (grey), with the discovered rest-leaning Φ schedule overlaid.

10 Discussion and open questions

The empirical results of Section 9 show that the SMC^2 filter and controller, composed in receding-horizon form, produce a closed-loop policy that improves mean autonomic amplitude by +17% over a sensible constant baseline on the FSA-v2 model, with no fatigue-ceiling violations.

Earlier drafts of these notes flagged three analyses as outstanding: Fisher information rank identifiability, Lyapunov stability, and a linear–quadratic–Gaussian (LQG) approximation. The first two have since been carried out and now form Sections 3 and 4. The third remains as the principal open piece of analytical work and is described in Section 10.3 below. Three smaller open questions are then collected in Section 10.4.

10.1 Resolved: Fisher information rank identifiability

The pen-and-paper FIM analysis of Section 3 identified three rank-deficient pairs in the original v2 parametrization, $(\kappa_B, \varepsilon_A)$, (μ_F, μ_{FF}), and (τ_F, λ_A). Each pair was shown to enter the one-day-window likelihood only through a single linear combination, which is the combinatorial signature of a rank-one rather than rank-two contribution to the FIM. The Stage G1 reparametrization rotates each pair into a strongly-identified combination and a residual that vanishes at the operating point of Definition 1; empirically this restores full FIM rank and contracts the posterior standard deviations on the residuals by roughly a factor of six (see Table 3).

A useful follow-up, identified in Remark 4, is to add a numerical FIM module under `smc2fc/diagnostics/fim/` that recomputes the rank and condition number directly from the JAX dynamics, so that any future modification of the model triggers an automatic re-check.

10.2 Resolved: Lyapunov stability of the SDE

Section 4 carried out the stability analysis in two parts. The deterministic skeleton, under a constant control $\Phi \equiv \Phi_c$, has an explicit $A = 0$ equilibrium whose local stability is determined entirely by the sign of the Stuart–Landau growth rate $\mu(B_0^*, F_0^*)$ (Proposition 5); the bifurcation locus of Section 4.1 crosses zero at $\Phi_c \approx 7.5$, above which overtraining-collapse renders the $A = 0$ branch stable. For the full SDE we exhibited a quadratic Lyapunov function $V = \frac{1}{2}(aB^2 + bF^2 + cA^2)$ that satisfies a Foster–Lyapunov drift inequality (Theorem 7), and concluded exponential ergodicity at rate $\rho < \alpha = \min(2/\tau_B, 1/\tau_F^{\text{old}})$ (Theorem 8). This delivers strong stability of the controlled process on the operating domain $\mathcal{X} = [0, 1] \times \mathbb{R}_+^2$ for any bounded control schedule.

Remark 9 noted that the constants in Theorem 7 are loose: they use the simplest quadratic Lyapunov function and global bounds on the residual factors. A sharper rate could be obtained from a piecewise Lyapunov function that switches between the $\mu < 0$ and $\mu > 0$ regimes identified in Proposition 5, or from a non-quadratic Lyapunov function tuned to the geometry of the Stuart–Landau cubic.

10.3 Outstanding: LQG / Riccati exact-approximate alternative

Section 8.8 sketched a linear–quadratic–Gaussian controller for FSA-v2 under three explicit assumptions:

(A1) linearisation of the drift around an operating point;

(A2) replacement of the state-dependent Jacobi/CIR diffusions by constant-coefficient Gaussian noise;

(A3) replacement of the asymmetric $-\int A\,\mathrm{d}t$ objective and the soft fatigue barrier by a quadratic surrogate cost.

Under (A1)–(A3), the optimal feedback law is the time-varying linear policy (36) with gain matrix $K(t) = R^{-1}B_{\mathrm{lin}}(t)^\top P(t)$, where $P(t)$ solves the differential Riccati equation (37). None of this is yet implemented in the repository.

We identify three concrete pieces of work that would deliver an LQG companion to the SMC^2 controller:

1. Implement automatic linearisation of the drift around a user-supplied operating point, using the existing JAX dynamics in `version_2/models/fsa_high_res/_dynamics.py`; expose the time-varying $A_{\mathrm{lin}}(t), B_{\mathrm{lin}}(t)$ of (34) on the project's fifteen-minute grid.

2. Implement a backward-integration solver for the differential Riccati equation (37).

3. Build an `LQGController` that emits the feedback gain $K(t)$ of (36) and a corresponding Φ schedule.

Two scientific questions then become accessible. *Does the LQG schedule, evaluated as an open-loop control on the FSA-v2 model, achieve mean amplitude comparable to the SMC^2 schedule on the* 84*-day benchmark of Section 9?* If yes, the non-linear SMC^2 machinery may be unnecessary in steady-state regimes; if no, the gap quantifies the value of capturing the cubic Stuart–Landau non-linearity. *Does using the LQG schedule as a warm-start for the SMC^2 control of Section 7 reduce wall-clock time to solution?* A reasonable hypothesis is that LQG provides an excellent initial θ_{ctrl} in the linear regime and only marginal benefit where the non-linearity matters; experimentally testing this is the natural next step.

10.4 Other open questions

A few smaller points deserve mention but are not, in the authors' view, on the same level of importance as the LQG analysis above.

- *Horizon length.* The Stage E5 result uses a fourteen-day macrocycle. The Stage D open-loop result uses eighty-four days. It is not yet known how the closed-loop gain scales with horizon length; a horizon sweep is straightforward to set up.

- *Posterior shape vs. posterior mean.* The controller of Section 8.4 uses only the posterior mean $\bar{\theta}_{\mathrm{ctrl}}$. The flat-cost-surface argument of Section 8.7 explains why this is sometimes sufficient, but in regimes where the cost surface is curved the full posterior carries useful information. A risk-sensitive variant that minimises a quantile of the cost rather than the mean would be the natural extension.

- *Parameter drift across windows.* The Bures–Wasserstein bridge of Section 6 produces a sensible Gaussian-fit warm-start, but does not protect against *slow* parameter drift in the truth (e.g. the athlete's time constants change with training age). A more principled approach would augment the static parameters with a slow random-walk component and let the filter track the drift explicitly.

With the FIM and Lyapunov analyses now in place, the model has both empirical validation through the Stage D and Stage E5 results of Section 9 and analytical guarantees on identifiability and stability. The remaining theoretical work concentrates on the LQG / Riccati alternative.

A Fisher information rank deficiencies of the pre-G1 model: derivations

This appendix supplies the derivations underlying the identifiability analysis of Section 3. We derive analytically the three rank-deficient parameter pairs of the pre-G1 FSA-v2 model (Sections A.1–A.3), describe the resulting null space of the FIM (Section A.4), and prove that the G1 reparametrization is mathematically equivalent to the pre-G1 model under a fixed relabelling of parameters (Section A.5).

Throughout, we use the operating point of Definition 1, write $\delta A = A - A_{\text{typ}}$ and $\delta F = F - F_{\text{typ}}$ for state fluctuations, and recall that at one-day windows the slow Banister state moves by at most $O(\mathrm{d}t/\tau)$ where $\tau \in \{\tau_B, \tau_F\}$, so δA and δF are small in this regime.

A.1 Pair 1: $(\kappa_B, \varepsilon_A)$

The pre-G1 B-drift is

$$\mathrm{d}B = \big[\kappa_B(1+\varepsilon_A A)\Phi - B/\tau_B\big]\mathrm{d}t + \cdots$$

Linearising the factor $1+\varepsilon_A A$ around A_{typ} and writing $\delta A = A - A_{\text{typ}}$,

$$\kappa_B(1+\varepsilon_A A)\,\Phi \;=\; \underbrace{\kappa_B(1+\varepsilon_A A_{\text{typ}})}_{=:\,\kappa_B^{\text{eff}}}\,\Phi \;+\; \kappa_B\,\varepsilon_A\,\delta A\,\Phi. \tag{38}$$

The first term carries the dependence of the daily drift on the parameters via the single product κ_B^{eff}; the second term is $O(\delta A)$, vanishing at the operating point.

Now consider how the data depend on small perturbations $(\delta\kappa_B, \delta\varepsilon_A)$. Differentiating (38) at the truth point gives

$$\frac{\partial}{\partial \kappa_B}\big[\kappa_B(1+\varepsilon_A A)\Phi\big] = (1+\varepsilon_A A)\,\Phi = (1+\varepsilon_A A_{\text{typ}})\,\Phi + \varepsilon_A \delta A\,\Phi,$$

$$\frac{\partial}{\partial \varepsilon_A}\big[\kappa_B(1+\varepsilon_A A)\Phi\big] = \kappa_B A\,\Phi = \kappa_B A_{\text{typ}}\,\Phi + \kappa_B \delta A\,\Phi.$$

At the operating point (where $\delta A = 0$) these two gradients are linearly dependent: their ratio is $(1+\varepsilon_A A_{\text{typ}})/(\kappa_B A_{\text{typ}})$, independent of Φ. The score function with respect to κ_B and the score function with respect to ε_A are therefore proportional, with proportionality constant fixed by the truth values. Their outer product in (12) is rank one, not rank two.

The combination that the data *can* resolve is precisely the product $\kappa_B^{\text{eff}} = \kappa_B(1+\varepsilon_A A_{\text{typ}})$. The orthogonal direction $(\delta\kappa_B, \delta\varepsilon_A) \perp \nabla\kappa_B^{\text{eff}}$ moves the parameter vector but leaves the daily drift unchanged at the operating point.

A.2 Pair 2: (μ_F, μ_{FF})

The pre-G1 Stuart–Landau growth rate is $\mu(B,F) = \mu_0 + \mu_B B - \mu_F F - \mu_{FF} F^2$. Completing the square in the quadratic in F,

$$\begin{aligned} -\mu_F F - \mu_{FF} F^2 &= -\mu_F F - \mu_{FF}\big[(F - F_{\text{typ}})^2 + 2F_{\text{typ}}F - F_{\text{typ}}^2\big] \\ &= -\big(\underbrace{\mu_F + 2F_{\text{typ}}\mu_{FF}}_{=:\,\mu_F^{\text{eff}}}\big)F \;-\; \mu_{FF}(F - F_{\text{typ}})^2 + \mu_{FF}F_{\text{typ}}^2. \end{aligned} \tag{39}$$

The constant $\mu_{FF}F_{\text{typ}}^2$ shifts the intercept μ_0 but is uninformative about μ_F or μ_{FF} separately; the linear-in-F slope μ_F^{eff} does carry their combined effect; and the quadratic $-\mu_{FF}(F - F_{\text{typ}})^2$ is $O(\delta F^2)$, so it contributes at second order in the small fluctuations of F at one-day windows.

Differentiating $\mu(B,F)$ with respect to μ_F and μ_{FF} at the truth point and setting $\delta F = 0$,

$$\partial_{\mu_F}\,\mu(B,F)\big|_{F_{\text{typ}}} = -F_{\text{typ}}, \qquad \partial_{\mu_{FF}}\,\mu(B,F)\big|_{F_{\text{typ}}} = -F_{\text{typ}}^2.$$

Both gradients point in the same one-dimensional subspace (they differ only by the scalar factor F_{typ}), so the FIM contribution from (μ_F, μ_{FF}) at the operating point is rank one. The combination resolved is $\mu_F^{\text{eff}} = \mu_F + 2F_{\text{typ}}\mu_{FF}$ (the slope at F_{typ}); the orthogonal direction is silent.

A.3 Pair 3: (τ_F, λ_A)

The pre-G1 F-clearance rate is $(1+\lambda_A A)/\tau_F$. Linearising in δA,

$$\frac{1+\lambda_A A}{\tau_F} = \frac{1+\lambda_A A_{\text{typ}}}{\tau_F} + \frac{\lambda_A}{\tau_F}\,\delta A =: \frac{1}{\tau_F^{\text{eff}}} + O(\delta A), \qquad \tau_F^{\text{eff}} := \frac{\tau_F}{1+\lambda_A A_{\text{typ}}}. \tag{40}$$

The leading term is the inverse effective acute timescale, set by the product $(1+\lambda_A A_{\text{typ}})/\tau_F$; the $O(\delta A)$ correction vanishes at the operating point. By the same argument as in Pair 1, the gradients of the F-decay rate with respect to τ_F and λ_A at the operating point are scalar multiples of one another, so the corresponding FIM contribution is rank one. The combination resolved is the inverse effective rate $1/\tau_F^{\text{eff}}$.

A.4 The null space of the FIM

Combining the three rank-one defects of Sections A.1–A.3, the FIM of the pre-G1 model evaluated at the operating point has a null space of dimension three. The three null vectors point along

$$(\delta\kappa_B, \delta\varepsilon_A) \perp \nabla\kappa_B^{\text{eff}}, \quad (\delta\mu_F, \delta\mu_{FF}) \perp \nabla\mu_F^{\text{eff}}, \quad (\delta\tau_F, \delta\lambda_A) \perp \nabla(1/\tau_F^{\text{eff}}), \tag{41}$$

i.e. along perturbations of each pair that hold the strongly identified linear combination fixed. The total FIM rank is therefore $d-3$, where d is the full dimension of the parameter vector.

The remaining $d-3$ identified directions include all observation-model parameters (which couple linearly to the latent state and are therefore generically identifiable from a one-day record under the sleep- and wake-gating structure of (8)–(11)), the slow Banister parameters τ_B and κ_F (which appear linearly and not in collinear pairs), and the diffusion scales $\sigma_B, \sigma_F, \sigma_A$ (which are identifiable from the empirical quadratic variation of the observation residuals).

A.5 The G1 reparametrization is mathematically equivalent to the pre-G1 model

Proposition 15 (G1 is mathematically equivalent to v2)**.** *The G1 drift* (3)–(5) *with the truth values of Table 1 is identically equal, as a function of* (B, F, A, Φ)*, to the pre-G1 drift under the relabelling*

$$\kappa_B^{\text{old}} = \kappa_B^{\text{new}}/(1+\varepsilon_A A_{\text{typ}}), \qquad \tau_F^{\text{old}} = \tau_F^{\text{new}}\,(1+\lambda_A A_{\text{typ}}),$$
$$\mu_F^{\text{old}} = \mu_F^{\text{new}} - 2F_{\text{typ}}\mu_{FF}, \qquad \mu_0^{\text{old}} = \mu_0^{\text{new}} - \mu_{FF}F_{\text{typ}}^2,$$

with $\varepsilon_A, \lambda_A, \mu_{FF}, \mu_B, \eta, \tau_B, \kappa_F$ *all unchanged.*

Proof. Substitute the relabelling into the pre-G1 drift terms (3)–(5) (in their original v2 form) and collect:

B-equation. The pre-G1 B-drift is $\kappa_B^{\text{old}}(1+\varepsilon_A A)\Phi - B/\tau_B$. Substituting $\kappa_B^{\text{old}} = \kappa_B^{\text{new}}/(1+\varepsilon_A A_{\text{typ}})$ gives $\kappa_B^{\text{new}} \cdot (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}}) \cdot \Phi - B/\tau_B$, which is exactly the G1 B-drift (3).

F-equation. The pre-G1 F-drift is $\kappa_F\Phi - (1+\lambda_A A)F/\tau_F^{\text{old}}$. Substituting $\tau_F^{\text{old}} = \tau_F^{\text{new}}(1+\lambda_A A_{\text{typ}})$ gives $\kappa_F\Phi - (1+\lambda_A A)/(1+\lambda_A A_{\text{typ}}) \cdot F/\tau_F^{\text{new}}$, which is exactly the G1 F-drift (4).

A-equation. The growth rate $\mu(B,F)$ becomes, under the relabelling,

$$\begin{aligned}\mu_0^{\text{old}} + \mu_B B - \mu_F^{\text{old}} F - \mu_{FF} F^2 &= (\mu_0^{\text{new}} - \mu_{FF} F_{\text{typ}}^2) + \mu_B B - (\mu_F^{\text{new}} - 2F_{\text{typ}}\mu_{FF})F - \mu_{FF}F^2 \\ &= \mu_0^{\text{new}} + \mu_B B - \mu_F^{\text{new}} F - \mu_{FF}(F^2 - 2F_{\text{typ}}F + F_{\text{typ}}^2) \\ &= \mu_0^{\text{new}} + \mu_B B - \mu_F^{\text{new}} F - \mu_{FF}(F - F_{\text{typ}})^2,\end{aligned}$$

which is exactly the G1 growth rate (2). The A-drift is then $\mu(B,F)A - \eta A^3$, identical in both parametrizations. □ □

The numerical truth values of Table 1 satisfy the above relabelling exactly, so the unit test `tests/test_g1_reparam.py` – which evaluates the two drift formulas at 50 random states – agrees to floating-point precision. The reparametrization is purely a coordinate change in parameter space; the dynamics, the observation model, and the joint likelihood $p(y_{1:T} \mid \theta)$ are unchanged.

B Stability of the FSA-v2 SDE: derivations

This appendix supplies the derivations underlying the stability analysis of Section 4. We compute the drift Jacobian that establishes Proposition 5 (Section B.1), set out the bifurcation locus explicitly (Section B.2), and prove the Foster–Lyapunov drift inequality of Theorem 7 (Section B.3).

B.1 The drift Jacobian at the $A = 0$ branch

Write the deterministic drift vector field as $f(B,F,A;\Phi_c) = (f_B, f_F, f_A)^\top$ with

$$\begin{aligned}f_B(B,F,A) &= \kappa_B\, g_B(A)\,\Phi_c - B/\tau_B, \\ f_F(B,F,A) &= \kappa_F\,\Phi_c - g_F(A)\,F/\tau_F, \\ f_A(B,F,A) &= \mu(B,F)\,A - \eta\,A^3,\end{aligned}$$

where $g_B(A) = (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}})$ and $g_F(A) = (1+\lambda_A A)/(1+\lambda_A A_{\text{typ}})$ are the residual factors. Their derivatives at $A = 0$ are $g_B'(0) = \varepsilon_A/(1+\varepsilon_A A_{\text{typ}})$ and $g_F'(0) = \lambda_A/(1+\lambda_A A_{\text{typ}})$.

The Jacobian $J = \partial f/\partial(B,F,A)$ at $(B_0^*, F_0^*, 0)$ has entries

- Row B: $\partial f_B/\partial B = -1/\tau_B$, $\partial f_B/\partial F = 0$, $\partial f_B/\partial A = \kappa_B g_B'(0)\Phi_c = \kappa_B\varepsilon_A\Phi_c/(1+\varepsilon_A A_{\text{typ}})$.
- Row F: $\partial f_F/\partial B = 0$, $\partial f_F/\partial F = -g_F(0)/\tau_F = -1/[\tau_F(1+\lambda_A A_{\text{typ}})]$, $\partial f_F/\partial A = -g_F'(0)F_0^*/\tau_F = -\lambda_A F_0^*/[\tau_F(1+\lambda_A A_{\text{typ}})]$.
- Row A: $\partial f_A/\partial B = \mu_B \cdot A = 0$ (at $A = 0$), $\partial f_A/\partial F = (-\mu_F - 2\mu_{FF}(F - F_{\text{typ}})) \cdot A = 0$ (at $A = 0$), $\partial f_A/\partial A = \mu(B,F) - 3\eta A^2 = \mu(B_0^*, F_0^*)$ (at $A = 0$).

In matrix form,

$$J(B_0^*, F_0^*, 0) = \begin{pmatrix} -1/\tau_B & 0 & \dfrac{\kappa_B\varepsilon_A\Phi_c}{1+\varepsilon_A A_{\text{typ}}} \\ 0 & -\dfrac{1}{\tau_F(1+\lambda_A A_{\text{typ}})} & -\dfrac{\lambda_A F_0^*}{\tau_F(1+\lambda_A A_{\text{typ}})} \\ 0 & 0 & \mu(B_0^*, F_0^*) \end{pmatrix}. \tag{42}$$

This is upper triangular, so its eigenvalues are the diagonal entries:

$$\lambda_1 = -1/\tau_B, \qquad \lambda_2 = -1/[\tau_F(1+\lambda_A A_{\text{typ}})], \qquad \lambda_3 = \mu(B_0^*, F_0^*). \tag{43}$$

The first two are unconditionally negative (with $\tau_B, \tau_F > 0$ and $\lambda_A, A_{\text{typ}} \geq 0$); the third has the sign of the Stuart–Landau growth rate. By Lyapunov's first method, the $A = 0$ equilibrium is locally asymptotically stable iff all three eigenvalues are negative, i.e. iff $\mu(B_0^*, F_0^*) < 0$. This proves Proposition 5. □

B.2 The bifurcation locus

The bifurcation locus $\{\Phi_c > 0 : \mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) = 0\}$ is the level set on which λ_3 in (43) vanishes. Substituting $B_0^*(\Phi_c) = \tilde{b}\,\Phi_c$ and $F_0^*(\Phi_c) = \tilde{f}\,\Phi_c$ with $\tilde{b} = \kappa_B\tau_B/(1+\varepsilon_A A_{\text{typ}})$ and $\tilde{f} = \kappa_F\tau_F(1+\lambda_A A_{\text{typ}})$ into the formula for μ,

$$\mu(B_0^*(\Phi_c), F_0^*(\Phi_c)) \;=\; \mu_0 + \mu_B\tilde{b}\,\Phi_c - \mu_F\tilde{f}\,\Phi_c - \mu_{FF}(\tilde{f}\Phi_c - F_{\text{typ}})^2. \tag{44}$$

This is a quadratic in Φ_c once expanded, with leading coefficient $-\mu_{FF}\tilde{f}^2 < 0$. At the truth values of Table 1,

$$\tilde{b} \;=\; \frac{0.01248\cdot 42}{1.04} \;\approx\; 0.504, \qquad \tilde{f} \;=\; 0.030\cdot 6.364\cdot 1.1 \;\approx\; 0.210,$$

giving $\mu(\Phi_c) = 0.036 + 0.0966\,\Phi_c - 0.40(0.210\,\Phi_c - 0.20)^2$. Solving $\mu(\Phi_c) = 0$ yields a single positive root $\Phi_c^{\text{crit}} \approx 7.5$ TRIMP-units per day.

For $0 < \Phi_c < \Phi_c^{\text{crit}}$ we have $\mu > 0$ and the $A = 0$ branch is unstable; for $\Phi_c > \Phi_c^{\text{crit}}$ we have $\mu < 0$ and the $A = 0$ branch is stable. At $\Phi_c = \Phi_c^{\text{crit}}$ the system undergoes a Hopf-like (super- or sub-critical depending on the sign of higher-order corrections we have not computed) bifurcation in the A-direction.

B.3 Foster–Lyapunov drift inequality (proof of Theorem 7)

We now prove that the quadratic Lyapunov function $V(B,F,A) = \frac{1}{2}(aB^2 + bF^2 + cA^2)$ of (15) satisfies the Foster–Lyapunov drift inequality (14) on the operating domain $\mathcal{X} = [0,1]\times[0,\infty)^2$, with rate $\alpha = \min(2/\tau_B,\, 1/[\tau_F(1+\lambda_A A_{\text{typ}})])$ and a finite constant $C > 0$.

Let $\mathcal{L}$ denote the infinitesimal generator of the SDE: for any $V \in C^2(\mathcal{X})$,

$$\mathcal{L}V(x) \;=\; \sum_{i\in\{B,F,A\}} b_i(x)\,\partial_i V(x) \;+\; \tfrac{1}{2}\sum_i a_i(x)\,\partial_{ii}^2 V(x), \tag{45}$$

with b_i the ith drift component and $a_B(x) = \sigma_B^2 B(1-B)$, $a_F(x) = \sigma_F^2 F$, $a_A(x) = \sigma_A^2 A$. We bound the three components separately. Throughout, $K_\bullet$ denotes a finite constant depending only on the truth parameters, $\Phi_{\max}$, and the choice of a, b, c.

B-component. Using $B \in [0,1]$ and the fact that $g_B(A) = (1+\varepsilon_A A)/(1+\varepsilon_A A_{\text{typ}})$ is bounded above on the operating domain by some $g_{B,\max} < \infty$ (depending on the practical upper bound on A that we wish to enforce; on the operating domain $A \in [0, A_{\max}]$ with any chosen $A_{\max}$, set $g_{B,\max} = (1+\varepsilon_A A_{\max})/(1+\varepsilon_A A_{\text{typ}})$),

$$\begin{aligned} aB\cdot\big[\kappa_B g_B(A)\Phi - B/\tau_B\big] + \tfrac{a}{2}\sigma_B^2 B(1-B) \;&\le\; a\,\kappa_B\,g_{B,\max}\,\Phi_{\max}\,B - (a/\tau_B)\,B^2 + \tfrac{a}{8}\sigma_B^2 \\ &\le\; a\,\kappa_B\,g_{B,\max}\,\Phi_{\max} + \tfrac{a}{8}\sigma_B^2 - (a/\tau_B)\,B^2, \end{aligned}$$

where the second line uses $B \le 1$ and the bound $B(1-B) \le 1/4$ on $[0,1]$. Define $K_B := a\kappa_B g_{B,\max}\Phi_{\max} + a\sigma_B^2/8$.

F-component. Using $g_F(A) \ge g_F(0) = 1/(1+\lambda_A A_{\text{typ}})$ for $A \ge 0$, and writing $\tau_F^{\text{old}} := \tau_F(1+\lambda_A A_{\text{typ}})$,

$$\begin{aligned} bF\cdot\big[\kappa_F\Phi - g_F(A)F/\tau_F\big] + \tfrac{b}{2}\sigma_F^2 F \;&\le\; b\,\kappa_F\,\Phi_{\max}\,F - bF^2/\tau_F^{\text{old}} + \tfrac{b}{2}\sigma_F^2 F \\ &=\; bF\,(\kappa_F\Phi_{\max} + \sigma_F^2/2) - bF^2/\tau_F^{\text{old}}. \end{aligned}$$

Apply Young's inequality $xy \le x^2/(2\beta) + \beta y^2/2$ to the linear-in-F term, with $\beta = \tau_F^{\text{old}}$ and $y = (\kappa_F\Phi_{\max} + \sigma_F^2/2)$:

$$bF(\kappa_F\Phi_{\max} + \sigma_F^2/2) \;\le\; bF^2/(2\tau_F^{\text{old}}) + (b\tau_F^{\text{old}}/2)(\kappa_F\Phi_{\max} + \sigma_F^2/2)^2.$$

Combined,

$$bF \cdot \left[\kappa_F \Phi - g_F(A)F/\tau_F\right] + \tfrac{b}{2}\sigma_F^2 F \;\le\; -(b/(2\tau_F^{\text{old}}))F^2 + K_F,$$

with $K_F := (b\tau_F^{\text{old}}/2)(\kappa_F \Phi_{\max} + \sigma_F^2/2)^2$.

A-component. The growth rate satisfies $\mu(B,F) \le \mu_{\max} := \mu_0 + \mu_B$ for $B \in [0,1]$ and $F \ge 0$, since the contributions $-\mu_F F$ and $-\mu_{FF}(F - F_{\text{typ}})^2$ are non-positive. Therefore

$$cA\big[\mu(B,F)\,A - \eta A^3\big] + \tfrac{c}{2}\sigma_A^2 A \;\le\; -c\eta A^4 + c\mu_{\max} A^2 + \tfrac{c}{2}\sigma_A^2 A.$$

Two applications of Young's inequality remove the unwanted A^2 and linear-A terms:

(i) For the A^4-vs-A^2 pair, complete the square with $x = A^2$:

$$-c\eta A^4 + c\mu_{\max} A^2 = -c\eta\big(A^2 - \mu_{\max}/(2\eta)\big)^2 + c\mu_{\max}^2/(4\eta) \;\le\; -\tfrac{c\eta}{2}A^4 + \tfrac{c\mu_{\max}^2}{2\eta},$$

where we discarded half of the negative quartic contribution to absorb the maximum of the non-negative quadratic.

(ii) For the linear A-term against the diffusion: by the elementary $A \le (\eta/(2\sigma_A^2))A^4 + \sigma_A^2/(2\eta)$ (setting $\alpha = \eta/(2\sigma_A^2)$ in $x \le \alpha x^4 + 1/(4\alpha)$ – proved by completing the square on $x \mapsto \alpha x^4 - x$ and verifying the minimum is $-1/(4\alpha)$, so $x - \alpha x^4 \le 1/(4\alpha)$, i.e. $x \le \alpha x^4 + 1/(4\alpha)$), and we then absorb the A^4 piece into a small fraction of the $-\frac{c\eta}{2}A^4$ term:

$$\tfrac{c}{2}\sigma_A^2 A \;\le\; \tfrac{c}{2}\sigma_A^2 \cdot \big(\eta/(2\sigma_A^2)A^4 + \sigma_A^2/(2\eta)\big) \;=\; \tfrac{c\eta}{4}A^4 + \tfrac{c\sigma_A^4}{4\eta}.$$

Wait – that produces the wrong sign on A^4. We instead want the linear-A term to sit on top of the cubic damping; redo with the opposite split. Use $A \le 1 + A^2$ (a strict bound for $A \ge 0$ verified by completing the square on $1 + A^2 - A$); then

$$\tfrac{c}{2}\sigma_A^2 A \;\le\; \tfrac{c\sigma_A^2}{2}(1 + A^2) \;=\; \tfrac{c\sigma_A^2}{2} + \tfrac{c\sigma_A^2}{2}A^2,$$

and the residual A^2 is then absorbed via the same complete-the-square argument as in step (i), but with $\mu_{\max}$ replaced by $\sigma_A^2/2$. Combining all three pieces,

$$cA[\mu(B,F)A - \eta A^3] + \tfrac{c}{2}\sigma_A^2 A \;\le\; -\tfrac{c\eta}{4}A^4 + \tfrac{c\mu_{\max}^2}{2\eta} + \tfrac{c\sigma_A^2}{2} + \tfrac{c\sigma_A^4}{8\eta} \;=:\; -\tfrac{c\eta}{4}A^4 + K_A.$$

Combining the three components. Adding the three bounds,

$$\mathcal{L}V(x) \;\le\; -(a/\tau_B)\,B^2 - (b/(2\tau_F^{\text{old}}))F^2 - (c\eta/4)\,A^4 + (K_B + K_F + K_A).$$

The first two negative terms already match $-\alpha V$ with $\alpha = \min(2/\tau_B,\, 1/\tau_F^{\text{old}})$. For the third we use the elementary inequality

$$-(c\eta/4)\,A^4 \;\le\; -(c\alpha/2)\,A^2 + c\alpha^2/(4\eta),$$

which follows by completing the square in A^2 on the parabola $(c\eta/4)x^2 - (c\alpha/2)x$. Plugging this in,

$$\mathcal{L}V(x) \;\le\; -\tfrac{\alpha}{2}\big(aB^2 + bF^2 + cA^2\big) + \big(K_B + K_F + K_A + c\alpha^2/(4\eta)\big) \;=\; -\alpha\,V(x) + C,$$

where $C := K_B + K_F + K_A + c\alpha^2/(4\eta)$ is finite. This is the required drift inequality (14), completing the proof of Theorem 7. □

The exponential ergodicity statement of Theorem 8 then follows from the standard Foster–Lyapunov ⇒ V-uniform ergodicity machinery in Khasminskii (1980) or, in discrete-time language transferable to continuous time via the resolvent, Meyn–Tweedie (1993, Theorem 6.1). The constants K_0 and ρ in (16) can be made explicit at the cost of further notation; we do not pursue this here.